

Maths of Charge and Electric Potential

[1] When a 5×10^{-9} test charge is placed at a point it experiences a force of $2 \times 10^{-4} \text{ N}$ in the x -direction that is electric field at the point.

Ans: we know that,

$$E = \frac{F}{q}$$

$$= \frac{2 \times 10^{-4}}{5 \times 10^{-9}}$$

$$= 4 \times 10^4 \text{ NC}^{-1}$$

Given that,

$$q = 5 \times 10^{-9} \text{ C}$$

$$F = 2 \times 10^{-4} \text{ N}$$

$$\text{Electric field } E = ?$$

[2] What is the force on an electron placed at the point where the electric field intensity is 4×10^4 .

we know that

$$E = \frac{F}{q}$$

$$\text{or, } F = qE$$

$$\text{or, } F = (4 \times 10^4 \text{ NC}^{-1})(-1.6 \times 10^{-19} \text{ C})$$

$$\text{or, } F = -6.4 \times 10^{-15} \text{ N}$$

Given that,

$$\text{Force } F = ?$$

$$E = 4 \times 10^4 \text{ NC}^{-1}$$

$$q = -1.6 \times 10^{-19} \text{ C}$$

3. Calculate the field E due to a dipole moment $4.5 \times 10^{-10} \text{ C/m}$ at a distance 1m from it on the perpendicular bisector.

Ans:

We know that,

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{p}{r^3}$$

$$= 9 \times 10^9 \times \frac{4.5 \times 10^{-10}}{1^3}$$

$$= 4.05 \text{ V/m}$$

Given that,

$$\text{Dipole moment, } p = 4.5 \times 10^{-10} \text{ C/m}$$

$$\text{Distance, } r = 1 \text{ m}$$

$$\text{Electric field, } E = ?$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

4. What is the potential at the surface of an aluminium nucleus $r = 4.5 \times 10^{-13} \text{ cm}$ and its charge $q = 13 \text{ C}$. (-1.6×10^{-19})

Ans:

We know that,

$$E = \frac{V}{r} \quad \text{--- (i)}$$

$$E = \frac{\sigma}{\epsilon_0} \quad \text{--- (ii)}$$

Equating (i) and (ii) $\Rightarrow$

$$\frac{V}{r} = \frac{\sigma}{\epsilon_0}$$

$$\text{or, } \frac{V}{r} = \frac{\phi}{\epsilon_0 4\pi r^2}$$

$$\text{or, } V = \frac{1}{4\pi\epsilon_0} \cdot \frac{\phi}{r}$$

$$\text{or, } V = (9 \times 10^9) \times \frac{2.08 \times 10^{-18}}{4.5 \times 10^{-15}}$$

$$\therefore V = 4.16 \times 10^6 \text{ V}$$

Given that,

$$r = 4.5 \times 10^{-15} \text{ m}$$

$$\text{charge, } q = 13 \times (-1.6 \times 10^{-19})$$

$$= -2.08 \times 10^{-18} \text{ C}$$

$$\text{potential, } V = ?$$

$$\text{charge density, } \sigma = \frac{\phi}{A} \quad \text{flux}$$

$$= \frac{\phi}{4\pi r^2}$$

$$V = \frac{\phi}{A} \cdot \frac{\phi}{4\pi r^2}$$

5 Two equal charges of magnitude $2 \times 10^{-6} \text{ C}$ are placed at a distance 8 cm from each other. Find the magnitude of electric intensity at point 3 cm from the midpoint of the line to the two charges along the perpendicular bisector of the line.

Ans: We know that,

$$\begin{aligned}
 E &= \frac{2aq}{4\pi\epsilon_0} \times \frac{1}{(a^2 + r^2)^{\frac{3}{2}}} \\
 &= \frac{1}{4\pi\epsilon_0} \times \frac{2aq}{(a^2 + r^2)^{\frac{3}{2}}} \\
 &= 9 \times 10^9 \times \frac{0.8 \times 2 \times 10^{-6}}{\{ (0.4)^2 + (0.3)^2 \}^{\frac{3}{2}}} \\
 &= 2 \times 10^9 \times 1.23 \times 10^{-5} \\
 &= 1.152 \times 10^5 \text{ N/C}
 \end{aligned}$$

Given that,

$$q = 2 \times 10^{-6} \text{ C}$$

$$2a = 8 \text{ cm} = 0.8 \text{ m}$$

$$a = 0.4 \text{ m}$$

$$r = 3 \text{ cm} = 0.3 \text{ m}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ NC}^{-1}\text{m}$$

$$E = ?$$

Q. An electric dipole consists of two opposite charges of magnitude $q = 2 \times 10^{-6} \text{ C}$ separated by 1 cm . The dipole is placed in an external field $2 \times 10^5 \text{ N/C}$. Calculate (i) the maximum torque on the dipole (ii) work done to turn the dipole through 180° . Starting from position at $\theta = 0^\circ$.

(i) We know that,

$$\begin{aligned}\tau &= pE \sin \theta \\ &= 2 \times 10^{-6} \times 2 \times 10^5 \times \sin 90^\circ \\ &= 4 \times 10^{-4} \text{ Nm}\end{aligned}$$

(ii) We know that,

$$\begin{aligned}W &= \int_0^{180} pE \sin \theta \cdot d\theta \\ &= -pE [\cos \theta]_0^{180} \\ &= -pE (\cos 180 - \cos 0) \\ &= -pE (-2) \\ &= 2pE \\ &= 2 \times 2 \times 10^{-6} \times 2 \times 10^5 \\ &= 8 \times 10^{-2} \text{ J}\end{aligned}$$

Given that

$$\begin{aligned}p &= q \times r = 2 \times 10^{-6} \times 10^{-2} \text{ m} \\ &= 2 \times 10^{-8} \text{ m}\end{aligned}$$

$$E = 2 \times 10^5 \text{ N/C}$$

Again,

$$p = 2 \times 10^{-8}$$

$$E = 2 \times 10^5 \text{ N/C}$$

$$W = ?$$

$$p = q \times r$$

$$= \frac{2 \times 10^{-6}}{100}$$

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$$E = 2 \times 10^5 \text{ N/C}$$

Again,

$$p = 2 \times 10^{-8}$$

$$E = 2 \times 10^5 \text{ N/C}$$

$$W = ?$$

$$= \frac{2 \times 10^{-8} \times 2 \times 10^5}{1}$$

Q. A 5 coulombs charge is placed on the circumference of a circle of radius 0.5m. Find the potential and intensity at a distance 25cm from the centre. $q = 5C$, $r = 0.5m$, $r' = 25cm$

(i) We know that,

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$= 9 \times 10^9 \times \frac{5}{0.5}$$

$$\therefore V = 9 \times 10^9 \text{ V}$$

Given that

$$q = 5C$$

$$r = 0.5m$$

$$\text{potential } V =$$

(ii) Electric field intensity

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

$$= 9 \times 10^9 \times \frac{5}{(25 \times 10^{-2})^2}$$

$$= 7.2 \times 10^{11} \text{ N/C}$$

$$q = 5C$$

$$r = 25 \times 10^{-2} m$$

$$E = ?$$

Q. What is the electric potential at the surface of a gold nucleus. The radius is 6.6×10^{-15} meter and the atomic number is 79.

Ans: We know that,

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$= 9 \times 10^9 \times \frac{1.264 \times 10^{-17}}{6.6 \times 10^{-15}}$$

$$= 1.723 \times 10^{-3} \text{ V}$$

Given that

$$r = 6.6 \times 10^{-15}$$

$$\text{Atomic number } (Z) = 79$$

$$q = ?$$

$$\text{charge, } q = Ze$$

$$= 79 \times 1.6 \times 10^{-19}$$

$$= 1.264 \times 10^{-17} \text{ coulomb}$$

$$V = ?$$

Q. At one time the positive charge in an atom was thought to be distributed uniformly throughout a sphere with a radius of about $1 \times 10^{-10} \text{ m}$. That is, throughout the entire atom calculate the electric field strength at the surface of a gold atom ($Z=79$) on this assumption.

Ans:

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

$$= 9 \times 10^9 \times \frac{1.264 \times 10^{-17}}{(1 \times 10^{-10})^2}$$

$$= 1.1376 \times 10^{13} \text{ N/C}$$

Given that,

$$r = 1 \times 10^{-10} \text{ m}$$

$$Z = 79$$

$$q = Ze = 1.264 \times 10^{-17} \text{ C}$$

$$E = ?$$

10] A negative point charge of 10^{-6} C is distributed in air of the origin of a rectangular co-ordinate system. A second negative point charge of 10^{-4} C is situated on the positive x-axis at a distance of 50 cm from the origin. What is the force on the second charge.

Ans:

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2} \times \hat{r}$$

$$= 9 \times 10^9 \times \frac{(-10^{-6})(-10^{-4})}{(50 \times 10^{-2})^2}$$

$$\Rightarrow F = 3.6 \text{ N}$$

$$\therefore \vec{F} = 3.6 \text{ N} (+\hat{i})$$

$$\left\{ \begin{array}{l} \text{Given that,} \\ q_1 = -10^{-6} \text{ C} \\ q_2 = -10^{-4} \text{ C} \\ r = 50 \times 10^{-2} \text{ m} \\ F = ? \end{array} \right.$$